USING EXTRATERRESTRIAL RESOURCES UNDER THE MOON AGREEMENT OF 1979

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Abstract: Capabilities of space launchers, space hardware and automation inexorably progress. Opportunities for use and exploitation of extraterrestrial resources approach practicality and affordability. Individuals and organizations interested in extraterrestrial resource use and exploitation have raised questions repeatedly about the legal status of resources beyond the Earth and the consequences of contemplated international controls. This article explores the relevant existing law, notes the vacuum of existing policies in the United States and in other space faring nations, notes relevant emerging organizational activity, and proposes possible avenues of approach. The unsuccessful attempt to address some extraterrestrial resource control questions through the 1979 Moon Agreement, must eventually be addressed squarely as a problem demanding solution. The near-term solution proposed in this article requires some courage and common sense, but does not propose entanglement in lengthy, inconclusive international negotiations. Progress depends upon innovative thinking and action, not on convoluted compromises of committees and councils. In this area, as in many other historically observable areas, law should follow, not precede the facts. Existing international law does not prohibit the use of space resources.

What Law is Applicable Today?

Considering the law of outer space as it applies to resources in outer space, we begin with the first major resolution of the United Nations (UN) General Assembly (GA) to deal with the status of resources in outer space, adopted 20 December 1961 in Resolution 1721 (XVI), the initial part of which (Paragraph A) reads:

> The General Assembly,
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> Recognizing the common interest of mankind in furthering the peaceful uses of outer space and the urgent need to strengthen international cooperation in this important field,
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> Believing that the exploration and use of outer space should only be for the betterment of mankind and to the benefit of States irrespective of the stage of their economic or scientific development,
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> 1. Commends to States for their guidance in the exploration and use of outer space the following principles:
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> (a) International law, including the Charter of the United Nations, applies to outer space and celestial bodies;

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(b) Outer space and celestial bodies are free for exploration and use by all States in conformity with international law and are not subject to national appropriation;

2. Invites the Committee on the Peaceful Uses of Outer Space to study and report on the legal problems which may arise from the exploration and use of outer space.

In the four subsequent paragraphs of this Resolution (B through E) there are no matters dealing directly with the status or use of extraterrestrial resources. The next significant UN action is contained in the *Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space,* adopted on 13 December 1963 in Resolution 1962 (XVIII), which states in its preambular provisions:

- the common interest of mankind in the progress of the exploration and use of outer space for peaceful purposes;
- the belief that the exploration and use of outer space should be carried on for the betterment of mankind and for the benefit of States;
- a desire to contribute to international cooperation in the scientific as well as in the legal aspects of the exploration and use of outer space;

and then the instrument proceeds with solemn declarations that, in the exploration and use of outer space, States should be guided by principles, including the following:

- The exploration and use of outer space should be carried on for the benefit and in the interests of all mankind.
- Outer space and celestial bodies are free for exploration and use by all States on a basis of equality and in accordance with international law.
- Outer space and celestial bodies are not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means.
- In the exploration and use of outer space, States shall be guided by the principles of cooperation and mutual assistance and shall conduct all their activities in outer space with due regard for the interests of other States.
- Whenever activities of one State cause harmful interference, or may be expected to cause harmful interference with the activities of another State in the exploration and use of outer space, consultation may be requested and should be conducted between or among the States involved concerning the activity(ies) in question.

Two important points emerge from a cursory study of these provisions: (1) the use of outer space by States, or by entities within States, is repeatedly declared as anticipated and allowed; and (2) there are no provisions denying the use of outer space. The only restrictions

declared have to do with the avoidance of harmful interference with activities in or uses of outer space by other States. In such a case, it is contemplated and urged that States involved would engage in consultations. These declarations are the most broadly accepted provisions of modern space law.[1] The 1963 Declaration of Principles laid a solid foundation in 1963 for the fundamental provisions of the later Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies.[2]

The 1967 Outer Space Treaty (OST)

This treaty was formulated within the UN Committee on the Peaceful Uses of Outer Space (COPUOS) and was adopted by the General Assembly (GA) of the UN and proffered to States for their signatures. In 1998 the OST had more than 100 signatures and more than 90 ratifications. All space faring States (*i.e.*, States capable of independently originating spaceflight) are ratified signatories to this treaty. Its provisions echo or amplify all the principles contained in the 1963 Declaration. As examples, the 1967 OST states in its Preamble that States Parties to the Treaty:

- Recognizing the common interest of all mankind in the progress of the exploration and use of outer space for peaceful purposes,
- Believing that the exploration and use of outer space should be carried on for the benefit of all peoples,
- Desiring to contribute to broad international cooperation in the scientific as well as the legal aspects of the exploration and use of outer space for peaceful purposes, and
- Recalling resolution 1962 (XVIII), which was adopted unanimously by the UNGA on 13 December 1963, and other former UN actions, agreed to a treaty which contains, among others, the following provisions.

Article I, paragraph 1: "The exploration and use of outer space, including the moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries, irrespective of their degree[s] of economic or scientific development, and shall be the province of all mankind." The second paragraph provides that "Outer space, including the moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies." The third paragraph reads "There shall be freedom of scientific investigation in outer space, including the moon and

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1 Having been unanimously adopted by the UNGA on 13 December 1963 in Resolution 1962 (XVIII).

2 18 UST 2410; TIAS 6347; 610 UNTS 205. Frequently referred to as the 1967 Outer Space Treaty, it was adopted by the UNGA on 19 December 1966 [Resolution 222 (XXI)], it was opened for signature on 27 January 1967 in London, Moscow and Washington. It entered into force 10 October 1967.

other celestial bodies, and States shall facilitate and encourage international cooperation in such investigation."

Article I does not say that any State may not use celestial bodies; conversely, it says that all States may. Access to and use of celestial bodies shall be free and equal. This is not to say that in order for State A to use a celestial body, it must provide for equal opportunity and means for such use by all other States; or that, in using a celestial body, State A must share all benefits of its use with all other States. Such extensions of meaning are illogical, unfounded and fly in the face of right reason and equity.

Article II provides that "Outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means." The fact that the use of a celestial body may not justify a declaration of national sovereignty over that body is not the same as saying "because one cannot declare sovereignty one cannot use." The OST says the opposite.

Every State is allowed to use the high seas, and no State is permitted to declare national sovereignty over the high seas. Similarly, every State has a right to use celestial bodies, but may not declare sovereignty over them. The inability to declare sovereignty is no constraint upon use.

Article III provides that "States Parties to the Treaty shall carry on activities in the exploration and use of outer space, including the moon and other celestial bodies, in accordance with international law, including the Charter of the United Nations, in the interest of maintaining international peace and security and promoting international cooperation and understanding." This is an unambiguous declaration of the applicability of international law, including the provisions of the Charter of the United Nations, to all that is done in outer space, including on the moon and other celestial bodies. It is also a clear exhortation to States to use outer space for peaceful purposes and, wherever practical and possible, to seek to promote international cooperation and understanding.

The promotion of cooperation and understanding with regard to commercial uses of outer space has been spectacularly successful in the application of satellite based communications, satellite navigation systems, and satellite remote sensing that now cover the world. International cooperative ventures of global and regional scope have emerged to establish means of using satellites in space applications through cooperation. There is no reason why cooperation could not be accomplished in the use of celestial bodies for activities such as propellant production, scientific exploration, establishment of settlements, or other ventures. There are no provisions of the 1967 OST, nor are there provisions in any other generally subscribed treaty relating to outer space that preclude uses of resources in outer space, including on the moon and other celestial bodies.

In Article IV, we find an explicit prohibition against placing in orbit around the Earth any object carrying nuclear weapons or any other weapons of mass destruction. Nor may such weapons be stationed in outer space in any manner, nor installed on celestial bodies. Article 4 declares

"The moon and other celestial bodies shall be used by all States Parties to the Treaty exclusively for peaceful purposes." Article IV further declares that "The establishment of military bases, installations and fortifications, the testing of any type of weapons and the conduct of military maneuvers on celestial bodies shall be forbidden." But, the Article also provides that: "The use of military personnel for scientific research or for any other peaceful purposes shall not be prohibited. The use of any equipment or facility necessary for peaceful exploration of the moon and other celestial bodies shall also not be prohibited." Thus the military use of outer space, including the moon and other celestial bodies, is explicitly constrained, but the uses of military personnel and equipment engaged in peaceful uses or scientific exploration are not prohibited. Again we find that the use of outer space, including the moon and other celestial bodies, is encouraged and facilitated for all peaceful purposes.

Article V, fully anticipating that different States Parties to the Treaty may be making simultaneous uses of outer space, including the moon and other celestial bodies, provides that "In carrying on activities in outer space and on celestial bodies, the astronauts of one State Party shall render all possible assistance to the astronauts of other States Parties."

Article VI clearly establishes a rule of State responsibility for the actions in outer space of the government or citizens of a State. "States Parties to the Treaty shall bear international responsibility for national activities in outer space [such as uses of outer space], including the moon and other celestial bodies, whether such activities are carried on by governmental agencies or by non-governmental entities, and [States are responsible] for assuring that national activities are carried out in conformity with the provisions set forth in the present Treaty." Similarly, "When activities are carried on in outer space, including the moon and other celestial bodies, by an international organization, responsibility for compliance with this Treaty shall be borne both by the international organization and by the States Parties to the Treaty participating in such organization." Article XIII of the 1967 OST also contains provisions relating to the responsibilities of international organizations.

Article VII establishes general rules relating to liability for injury to third parties resulting from the conduct of spaceflight activities. This article is the basis for an elaborated international convention to deal with the liability issues related to damages caused by spaceflight activities. See the 1972 *Convention on International Liability for Damages Caused by Space Objects* (the 1972 Liability Convention).[3]

Article VIII clarifies and elaborates aspects of the ownership of objects launched into outer space and their status in outer space, and establishes an obligation for States Parties to the Treaty who may discover a returned object on their territory to return such object to the State of

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[3] 24 UST 2389; TIAS 7762; 961 UNTS 187. This convention was negotiated through the UNCOPUOS and was adopted by the UNGA in Resolution 2777 (XXVI) on 29 November 1971. Opened for signature on 29 March 1972 in London, Moscow and Washington; it entered into force on 1 September 1972.

origin. An elaborated agreement was established subsequently to provide for return of astronauts and space objects that return to Earth. See the 1968 *Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched Into Outer Space* (1968 Rescue Agreement).[4]

In Article IX we have an elaboration of the duty of care and State responsibility for activities conducted in space, as well as a duty to communicate matters of potential harm. Article X provides for a right of reciprocal visits between sites of launch operations, subject to prior agreement between the States involved. Article XI establishes a duty to report on activities undertaken in outer space. Article XII, like Article X, provides a reciprocal right of visitation between space installations, subject to prior coordination between the States involved. Article XIII explicitly provides that the OST applies to activities of States Parties in outer space, whether they are undertaken singly or through international organizations. Articles XIV through XVII contain the housekeeping provisions relating to signature, ratification, depository, entry into force, amendment and other procedural aspects of the treaty.

In its total impression, the 1967 OST is an enabling treaty that formally establishes and declares a body of relevant law that applies to the exploration and the use of outer space, including the moon and other celestial bodies. With the exception of prohibitions related to defined military uses of outer space, the treaty is promotional and enabling, it is not restrictive. It even encourages visits between installations in space, subject to prior agreement between the parties involved (see Article 12).

It is similarly clear that the ancillary treaties that followed the 1967 OST, namely the 1968 Rescue and Return Agreement, the 1972 Liability Convention, and the 1975 Registration Convention[5] dealt with the prospects of exploration and use of outer space, and all their provisions are consistent with an expectation of such exploration and use.

The 1979 Moon Agreement

In contrast to the foregoing discussion and provisions of the aforementioned treaties and agreements, the 1979 Moon Agreement[6] is fundamentally different in nature. Because it is regarded by critics as

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[4] 19 UST 7570; TIAS 6599; 672 UNTS 119. This agreement was negotiated through the UNCOPUOS and was adopted by the UNGA in Resolution 2345 (XXII) on 19 December 1967. The agreement was opened for signature on 22 April 1968 in London, Moscow and Washington, and entered into force on 3 December 1968.

[5] *Convention on Registration of Objects Launched into Outer Space* (Registration Convention) 28 UST 695; TIAS 8480; 1023 UNTS 15. The convention was negotiated through the UNCOPUOS and was adopted by the UNGA on 12 November 1974 in Resolution 3235 (XXIX). The convention was opened for signature in New York on 14 January 1975 and entered into force on 15 September 1976.

[6] *Agreement Governing the Activities of States on the Moon and Other Celestial Bodies* (Moon Agreement) 18 INT'L LEGAL MAT'LS. 1434; 1363 UNTS 3. This treaty was negotiated through the UNCOPUOS and was adopted by the UNGA on 5

restrictive, the agreement has languished for want of signatures and ratifications. It is the least accepted agreement of the treaties negotiated through the UNCOPUOS. The 1979 Moon Agreement has fewer than twenty signatories and fewer than 10 ratifications. Major space faring nations, with the exception of France, are not signatories and are not likely to become signatories. The crux of the problem with the Moon Agreement is embodied in its Article 11. We examine Article 11 in some detail below, but let us first consider the earlier elements of the treaty. The agreement is flawed, not only in its concept and design, but also in that it proposes to establish new international law inconsistent with existing and widely accepted principles of law.

In its Preamble, the 1979 Moon Agreement provides that the Parties Signatory:

"- Noting the achievements of States in the exploration and use of the moon and other celestial bodies, [there having been by this time successful landings of men on the moon and return of material specimens to Earth from manned and automated flights to the moon;]

"- Recognizing that the moon, as a natural satellite of the earth, has an important role to play in the exploration of outer space,

"- Determined to promote on the basis of equality the further development of cooperation among States in the exploration and use of the moon and other celestial bodies, [noting again that "use" is explicitly contemplated]

"- Desiring to prevent the moon from becoming an area of international conflict,

"- Bearing in mind the benefits which may be derived from the exploitation of the natural resources of the moon and other celestial bodies," [now explicitly extending the notion of "use" to embrace "exploitation"]

"- [then] recalling the 1967 Outer Space Treaty, the 1968 Rescue and Return Agreement, the 1972 Liability Convention, and the 1975 Registration Convention, and

"- Taking into account the need to define and develop the provisions of these international instruments in relation to the moon and other celestial bodies, having regard to further progress in the exploration and use of outer space,

"Have agreed on the following:"

The Preamble clearly indicates the drafters' intention to define and develop the previously agreed space treaties in relation to the moon and other celestial bodies. It is the case that substantial portions of the Moon Agreement restate many of the provisions of earlier agreements with little or no definition or amplification. In the following excerpts we con-

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December 1979 in Resolution 34/68. The Moon Agreement was opened for signature on 18 December 1979 in New York and entered into force on 11 July 1984.

centrate on those provisions of the Moon Agreement that contain new materials, agreements or definitions, not already found in the former agreements. We do not quote or dwell on obvious declarations and repetitive provisions. In two material elements, the 1979 Moon Agreement proposes to change existing, well established principles of international law.

Article 1 provides definitionally that the Moon Agreement applies to the celestial bodies within the solar system, other than the Earth, except in so far as specific legal norms enter into force with respect to any of these [solar system] celestial bodies. This is a declaration limiting applicability of the agreement to the bodies of the solar system, other than the Earth. For purposes of this Agreement, reference to the moon includes orbits around or other trajectories to or around the moon. Also, the Moon Agreement declares that it does not apply to extraterrestrial materials which reach the surface of the Earth naturally.

Article 2, in its entirety, is one of the provisions which might well have been excluded as totally redundant and superfluous. Article 2 neither adds nor states anything new.

Article 3 is totally redundant in the first paragraph. The second paragraph is a restatement of the provisions of the UN Charter concerning the threat or use of force, which was already applicable to the moon and other celestial bodies by virtue of Article III of the 1967 OST. The third and fourth paragraphs are an unnecessary restatement of the consequences of Article IV of the 1967 OST. Thus, Article 3 of the Moon Agreement neither adds nor states anything new.

Article 4, paragraph 1, reads: "1. The exploration and use of the moon shall be the province of all mankind and shall be carried out for the benefit and in the interests of all countries, irrespective of their degree[s] of economic or scientific development. Due regard shall be paid to the interests of present and future generations as well as to the need to promote higher standards of living and conditions of economic and social progress and development in accordance with the Charter of the United Nations." This paragraph is, in part, a restatement of the 1963 Declaration of Principles and the first paragraph of Article I of the 1967 OST, and, in part [the second sentence] so general and vague as to be irrelevant. This is a provision reflecting the interests of States that believe they are not sufficiently represented among the space faring nations of the world, fearing that they may be left out or ignored in the process of possible future exploitation of lunar resources.

A far more reliable guarantee of participation by such countries would be obtained by their development of national competence in selected or specialized space technology areas, such as we see being done in India, Brazil and selected other developing countries, and the involvement of their citizens or institutions in international cooperative ventures to visit and use the resources of the moon and other celestial bodies.

The models of global participation in and use of satellite communication, navigation, remote sensing and meteorological services come to mind. States cannot enjoy the benefits of satellite communications

or other space application services unless they equip themselves with resources to do so. The same observation can be made about use and exploitation of lunar resources. States interested in participating should equip themselves with the resources to do so.

Article 4, paragraph 2, reads: "2. States Parties shall be guided by the principle of cooperation and mutual assistance in all their activities concerning the exploration and use of the moon. International cooperation in pursuance of this Agreement should be as wide as possible and may take place on a multilateral basis, on a bilateral basis or through international intergovernmental organizations." A statement of the obvious. This paragraph of Article 4 is a similarly ineffective attempt to "protect" States' perceived rights, which would be better accomplished through direct, sustained and equitable participation in the actual missions to be undertaken. When and how shall the peoples of the world know that the "international cooperation pursuant to this Agreement ...[is] as wide as possible..."? Who shall make such a determination?

Article 5 is an attempt to require any State undertaking a moon mission of any kind to fully disclose every operational aspect and every significant discovery or result of such a mission. Updates on information "shall be given periodically" at prescribed intervals. This is a new and unprecedented requirement of "full disclosure." The notification provisions of Article 5, paragraph 2, are totally redundant to good sense and protection of self-interest. Article 5, paragraphs 1, 2 and 3, are elaborations of detail of the obligations already imposed by provisions of Article IX of the 1967 OST. While they extend the detail of implications, these paragraphs create no new obligations.

Article 6, paragraph 1, is fully redundant to provisions of the 1967 treaty: "1. There shall be freedom of scientific investigation on the moon by all States Parties without discrimination of any kind, on the basis of equality and in accordance with international law." One is tempted to ask, "Is discrimination against non-signatory States allowed?" Paragraph 2 provides that "In carrying out scientific investigations and in furtherance of the provisions of this Agreement, the States Parties shall have the right to collect on and remove from the moon samples of its mineral and other substances." This had already been done during the 1970's by the United States and the then USSR.

The second paragraph continues "Such samples shall remain at the disposal of those States Parties which caused them to be collected and may be used by them for scientific purposes. States Parties shall have regard to the desirability of making a portion of such samples available to other interested States Parties and the international scientific community for scientific investigation. *States Parties may in the course of scientific investigations also use minerals and other substances of the moon in quantities appropriate for the support of their missions.*" (Italics added.) Thus, there is explicit authority in the 1979 Moon Agreement to "use mineral and other substances of the moon in quantities appropriate for the support of ... missions." The extent of the use of materials "in support of a

mission" is then determinable by the mission definition or mission profile, defining its purposes and objectives.

Paragraph 3 of Article 6 of the Moon Agreement provides that "States Parties agree on the desirability of exchanging scientific and other personnel on expeditions to or installations on the moon to the greatest extent feasible and practicable." Again, it is obviously contemplated that there will be missions to the moon and installations established there, with personnel who should be exchanged between installations and missions "to the greatest extent feasible and practicable."

Article 7 is devoted to the preservation of the environment on the moon. States Parties "shall take measures to prevent the disruption of the existing balance of the environment" and they shall "avoid harmfully affecting the environment of the [E]arth through the introduction of extraterrestrial matter or otherwise." Paragraph 2 requires prior notice to the UN Secretary-General of any intention to introduce nuclear materials on the moon. In paragraph 3, the Agreement requires notice to the UN Secretary-General of the need for any special scientific preserves for which protective arrangements should be agreed.

Article 8 permits landing on and launching space objects from the surface of the moon; placement of personnel, space vehicles, equipment, facilities, stations, and installations anywhere on or below the surface of the moon; and all these people and materials may move or be moved freely over or below the surface of the moon, subject to the avoidance of any harmful interference with personnel or property of another State Party present on the moon. If such harmful interference is possible, there is a duty to consult between or among the involved or interested States. The agreement is silent with regard to States relations with settlements that are independent of States Parties and thus have an existence outside the agreements among States Parties.[7]

Article 9 permits the establishment of manned or unmanned stations on the moon. The establishing State(s) "shall only use that area which is required for the needs of the station and shall immediately inform the Secretary-General of the United Nations of the location and purposes of that station", as well as providing an annual update on the use and purposes of the station.

Article 10 of the Moon Agreement says that Article V of the 1967 OST shall apply to persons on the moon and that the 1975 Rescue and Return Agreement shall be relevant. Article V of the 1967 treaty, as well as the 1975 Agreement, would apply to persons on the moon regardless of the content of Article 10. This is another article fully redundant to prior agreements. So much of what is in the Moon Agreement is repetitive of what is in the earlier, much more broadly subscribed agreements. It is clear that the main problem that creates the general unacceptability of the Moon Agreement lies in Article 11.

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[7] See discussion of "autonomy" in Sterns, Stine and Tennen, source at note 8, at 59-60.

In Article 11, paragraph 1, reads: "1. The moon and its natural resources are the common heritage of mankind, which finds its expression in the provisions of this Agreement, in particular in paragraph 5 of this article." Gobbledygook! The declaration that "the moon and its natural resources are the common heritage of mankind," is a phrase burdened with a history of inconsistent interpretations about the meaning of the term "common heritage of mankind," and what that term connotes with regard to the exploitation or use of lunar resources.

Article 11, paragraph 2, reads: "2. The moon is not subject to national appropriation by any claim of sovereignty, by means or use or occupation, or by any other means." This is a painfully obvious restatement of the content of Article II of the 1967 OST and, like many other parts of the Moon Agreement, is a paragraph that is totally redundant and unnecessary.

Article 11, paragraph 3, reads:

> 3. Neither the surface nor the subsurface of the moon, nor any part thereof or natural resources in place, shall become property of any State, international intergovernmental or non-governmental organization, national organization or non-governmental entity or of any natural person. The placement of personnel, space vehicles, equipment, facilities, stations and installations on or below the surface of the moon, including structures connected with its surface or subsurface, shall not create a right of ownership over the surface or the subsurface of the moon or any areas thereof. The foregoing provisions are without prejudice to the international regime referred to in paragraph 5 of this article.

This provision appears to declare a moratorium on extending property rights to materials "in place" on the moon, pending the establishment of a regime of resource management by means described in paragraph 5 of Article 11. The intention was to withhold property rights over materials in place on the moon, but not to constrain use of lunar materials.[8] While materials are "in place," they are *res communes,* the property of everyone. Once material is reduced to use, this point is moot. To argue that there is a moratorium contained in the Moon Agreement on

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[8] See the diverse discussions of the legal nature and history of this matter in Fasan, E. *Dominium Lunae, Proprietas Lunae;* Cocca A. A., *Property Rights on the Moon and Other Celestial Bodies;* Almond, H. H., *The Legal Status of Property on the Moon and Other Celestial Bodies;* Mani, V. S., *The Common Heritage of Mankind: Implications for the Legal Status of Property Rights on the Moon and Other Celestial Bodies;* van Traa-Engelman, H. L., *Clearness Regarding Property Rights on the Moon and Other Celestial Bodies*; Gál, G., *Acquisition of Property in the Legal Regime of Celestial Bodies;* and Sterns, P. M., G. H. Stine, and L. I. Tennen, *Preliminary Jurisdictional Observations Concerning Property Rights on the Moon and Other Celestial Bodies in the Commercial Space Age,* 39 PROC. COLLOQ. L. OUTER SPACE 1-60 (1997).

use of lunar material, one must ignore the explicit provisions of Article 6, paragraph 2, which states that: "States parties may... use mineral and other substances of the moon in quantities appropriate for the support of their missions."

Consider the air you breathe. You do not own the air. It is *res communes*. But you breathe it. When you breathe it, you extract oxygen and add carbon dioxide and then exhale it. Once exhaled, the air is again *res communes*, until inspired again. You don't own the air, but you certainly make extensive use of it. The same may be said of sunlight. No one owns the sunlight, but many people use it. Some use it to produce energy for power applications, both in space and on the Earth. Farmers use sunlight to grow crops. Some use the sun to heat water, or their homes, but no one owns sunlight. What are the consequences to the users of not being allowed to own the sunlight? The consequences are of no effect. One could argue that the provisions of Article 11, paragraph 3, of the Moon Agreement have about the same force, which is to say, no force at all, with regard to use of lunar materials.

Paragraph 4 of Article 11 reads: "States Parties have the right to exploration and use of the moon without discrimination of any kind, on the basis of equality and in accordance with international law and the terms of this Agreement." The most constructive aspect of this largely redundant provision is that it makes clear that Article 11 of the Moon Agreement was not at the time of its drafting, and is not now a part of international law, except as to relationships between the Parties to the Agreement. This rewrite of the second paragraph of Article I of the 1967 OST does not create any new international law.

Paragraph 5 of Article 11 reads: "States Parties to this Agreement hereby undertake to establish an international regime, including appropriate procedures to govern the exploitation of the natural resources of the moon as such exploitation is about to become feasible. This provision shall be implemented in accordance with Article 18 of this Agreement."

Thus the States Parties to the Agreement have declared an undertaking, but as to States non-Parties to the Agreement, there is no such undertaking nor are they bound by what the States Parties choose to do, especially in the case where the States Parties comprise fewer than ten percent of the world's nations.

Among the most offensive and inconsistent provisions of the Moon Agreement are those embodied in the seventh paragraph of Article 11, which reads:

> The main purposes of the international regime to be established shall include:
> (a) The orderly and safe development of the natural resources of the moon;
> (b) The rational management of those resources;
> (c) The expansion of opportunities in the use of those resources; [and]

(d) An equitable sharing by all States parties in the benefits derived from those resources, whereby the interests and needs of the developing countries, as well as the efforts of those countries which have contributed either directly or indirectly to the exploration of the moon, shall be given special consideration.

Paragraphs 11 (a) through 11 (c) are unobjectionable statements based in common sense. Paragraph 11 (d) however, contradicts paragraph 11 (b) and contradicts Article 4, paragraph 1 of the Moon Agreement, Article I, paragraph 1, of the 1967 OST, and provisions of early UN Resolutions on principles adopted by the UNGA.

Why and how are the interests and needs of developing countries to be given special consideration with regard to the sharing of benefits derived from the use or exploitation of lunar resources? The language of Article 11, paragraph 7 (d) is in direct contradiction of the language of Article I, paragraph 1, of the 1967 OST and of Article 4, paragraph 1, of the Moon Agreement. Either we should carry out activities in outer space, including the moon and other celestial bodies, "in the interests of all countries irrespective of their degree[s] of economic and scientific development," or we will share the benefits derived from lunar resources giving special consideration to "the interests and needs of the developing countries" and those which have contributed to the effort. We can do one or the other, but we cannot do both at the same time. Because Article 11, paragraph 7 (d) is only a hortatory declaration of future intent, it must fail, to the extent it conflicts with the language of earlier UN Resolutions, the language of the 1967 OST, and the language of Article 4 of the Moon Agreement. Paragraph 7 (d) is just bad policy, whether current or future, inconsistent with established law, and ought to be ignored rather than implemented.

Article 11 concludes with paragraph 8, which reads: "All the activities with respect to the natural resources of the moon shall be carried out in a manner compatible with the purposes specified in paragraph 7 of this article and the provisions of article 6, paragraph 2, of this Agreement." Compliance with paragraph 7 (d) would be action in violation of broadly accepted, well established international law.

Article 12, paragraph 1, is a redundant restatement of provisions contained in Article VIII of the 1967 OST. Article 12 Paragraph 2, is an unnecessary declaration that the 1975 Agreement on Rescue and Return of astronauts and space objects applies on the moon. Article 12, paragraph 3 states that "In the event of an emergency involving a threat to human life, States Parties may use the equipment, vehicles, installations, facilities or supplies of other States Parties on the moon. Prompt notification of such use shall be made to the Secretary-General of the United Nations or the State Party concerned." This provision, being devoid of constraints of reasonableness and prior notice, has potential for possibly life-threatening interference, with potential counter productive expansion of the emergency to involve others. The provision has many implications which appear not to have been thought through very well. By eliminating

the requirement of prior notice, as required by Article XII of the 1967 OST for reasons stated there, this provision in Article 12, paragraph 3, of the Moon Agreement is a potentially dangerous and an undesirable provision that ought not to survive any international review of the Moon Agreement. As it stands, Article 12, paragraph 3 of the 1979 Moon Agreement, invalidates the notice requirements of Article XII of the 1967 OST.

Article 13 would appear to be another logical extension of the obligations of Article VIII of the 1967 OST.

Article 14, paragraph 1, is a wholly redundant restatement of provisions found in Article VI of the 1967 OST. The second paragraph of Article 14 is another masterful restatement of the obvious.

Article 15 is a provision establishing a right on the part of any State Party to the Agreement to visit the space vehicles, equipment, facilities, stations, and installations of any other State Party on the moon. It is a sort of "general warrant" that permits anyone who gets to the Moon to gain access to whatever it is that anyone else on the Moon is doing there. This provision creates a people's police force by permitting unlimited inspection after the giving of reasonable notice of a planned visit. The bulk of this longest article in the agreement concerns itself with methods and procedures for the resolution of differences and disputes that may arise between or among States Parties to the Agreement. It is an Article calculated to give rise to many disagreements and disputes.

Article 16 extends general application of the terms of the Agreement to the activities of consenting international intergovernmental organizations, if a majority of the members of the organizations are also signatories to the Moon Agreement and the 1967 OST.

Articles 17 through 21 deal with the procedural aspects of the Agreement. Provisions in Article 18 provide for a review of the Agreement after 10 years,[9] which took place and resulted in no change. Article 18 also permits the UN Secretary-General to convene a review conference at the request of one third of the States Parties to the Agreement, with the concurrence of a majority of the States Parties. No such conference has been requested or convened.

Other Applicable Law

Beyond the treaties formulated in the UNCOPUOS relating to activities of States in the exploration and use of outer space, there is no presently identifiable law relating to the use or exploitation of resources on the moon, or for that matter, on any other celestial body, including asteroids.

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[9] According to E. Fasan, on 9 December 1994 the UNGA decided not to take action regarding revision of the Moon Agreement, although the ten year period since its entry into force had elapsed. Fasan, E., *Dominium Lunae, Proprietas Lunae*, 39 PROC. COLLOQ. L. OUTER SPACE 1 (1997). *See* UNGA Resolution 49/34 dated 9 December 1994.

Although there have been several national bodies of space law established, none of them to date has dealt explicitly with the legal status of or regulations relating to resources in outer space. There is no identifiable, relevant law or policy in the United States today, beyond the generally applicable provisions of the 1967 OST. At this writing the author is unaware of any national law relating to the status, use or exploitation of extraterrestrial resources.

Alternative Approaches Emerging

There is a program being developed by the recently formed United Societies in Space, Inc., in Denver, Colorado.[10] This group is formulating plans for a Lunar Economic Development Authority.[11] The proposed authority would be organized through international cooperation and operated as a licensing body to plan, coordinate, facilitate and encourage commercial exploitation of the resources of the Moon. The concept is not intended to apply only to the Moon, because other, wider ranging ventures may be undertaken in the future. At this time, however, the organizers believe that the Moon is the most likely early site of the actual use of extraterrestrial resources. The initial focus is on creating a Lunar Economic Development Authority. When technology and economics permit a wider range of activities off the Earth, the concept would be flexible enough to accommodate broadening the scope of activities or creating an alternative entity to oversee activities other than those related to the Moon.

Aldo Armando Cocca argues that a new international agency is the best potential means for accomplishing the harmonization of the interests of mankind as a whole with the interests of States or entities arriving on and seeking to exploit the resources of the Moon.[12] Harry Almond argues that substantial studies, analyses and discussions are required to understand what it is in outer space that is intended to be regulated and how it is intended to regulate or control activities. His analysis is influenced by perceptions of power and use of power to establish and protect rights in property. Almond believes that there is a great deal of ambiguity and uncertainty related to the expectations of States with regard to activities, including exploitation of resources, in outer space. He offers a list of proposed courses of action to reduce the ambiguities and uncertainties.[13]

V. S. Mani suggests that there is a need for an International Space Resource Management Authority with close linkage to the United Nations,

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[10] For information, contact Mr. Declan O'Donnell, United Societies for Space, Inc., 3300 East Fourteenth Avenue, Denver, CO 80206 USA.

[11] O'Donnell, D. J., *Benefit Sharing: The Municipal Model,* 39 PROC. COLLOQ. L. OUTER SPACE 151-161 (1997).

[12] Cocca A. A., *Property Rights on the Moon and Other Celestial Bodies,* 39 PROC. COLLOQ. L. OUTER SPACE 9-19 (1997)

[13] Almond, H. H., *The Legal Status of Property on the Moon and Other Celestial Bodies,* 39 PROC. COLLOQ. L. OUTER SPACE 20-30 (1997).

associated with numerous other activities deemed necessary for proper progress to be achieved. Mani argues that the concept of the "common heritage of mankind" is governing in this matter and concludes a presentation of his highly biased views with this question:

"Will the Space Powers rise above their short term, narrow, Shylockian view of profits, and meet the challenge of the next millennium with magnanimity, compassion and camaraderie to the whole international community?"[14]

One may hope. One may hope that the Space Powers will go right on exploiting the world by promoting, creating and supporting things like: a United Nations Organization; global attacks on diseases like smallpox, poliomyelitis, cancer and other scourges; and dozens of peace-keeping teams deployed over the Earth. One may hope that they will continue providing technologies for real-time communication capability for every person on Earth; and global real-time and reliable navigation; global weather monitoring and services such as the Internet. When the critics of personal freedom and competition can view reality with objectivity, greater progress will be made. Discord, name-calling and unrestrained covetousness do not foster progress, they hamper it.

H.L. van Traa-Engelman approaches this entire subject area in a similarly unrealistic manner to that of Mani. Treating the Agreement on the Moon as if it represents the global community's accepted approach, and is thus the prevailing international law, will not result in the acceptance of the proposed approach. The major difficulty in the approach of van Traa-Engelman and others is analogous to trying to mix oil with water. Consider this suggestion:

"As soon as the international regime will be established in accordance with Art. 11 of the Moon Treaty, States Parties will have to make provisions to honour specific property rights on natural resources of the moon etc. acquired through national legislation, but to be accommodated within the constraints and requirements of such an international regime."[15]

This declaration ignores the apparent and declared intent of the proposed international regime, which was and is to negate individual property rights in the Moon's resources and to establish the rights of mankind in equality, equity and with special consideration to the interests and the needs of developing countries and those contributing to the exploration of the Moon. Gyula Gál clearly states the problem with this approach:

"Unfortunately mankind is not a subject of international law. It can not act as a legal person, can not launch a process, can not claim anything in its own name. ... Coming generations have not a single 'ombudsman'

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[14] Mani, V. S., *The Common Heritage of Mankind: Implications for the Legal Status of Property Rights on the Moon and Other Celestial Bodies*, 39 PROC. COLLOQ. L. OUTER SPACE 31, at 36 (1997).

[15] van Traa-Engelman, H. L., *Clearness Regarding Property Rights on the Moon and Other Celestial Bodies*, 39 PROC. COLLOQ. L. OUTER SPACE 43 (1997).

among us to represent their interests, which we cannot understand in many respects. For our subject [Acquisition of Property in the Legal Regime of Heavenly Bodies]: it would be science fiction to speak about property rights on heavenly bodies which some time in the future will be acquired by mankind itself."[16]

It may be that Mani's "overwhelming majority of the contemporary international community" desires such an outcome. It is difficult to see how the resources necessary to permit exploitation of lunar resources would ever be amassed, organized and brought to bear on the problem, if exploitation is to be under an international administration, on behalf of "mankind," the first premise of which is "there are no property rights."

Conclusion

The 1979 Moon Agreement is an imperfect, largely redundant, partisan instrument, attempted to be imposed upon the space faring States by others. The ultimate defense against such imposition is by States declining to become signatories. The 1979 Moon Agreement was badly conceived and poorly drafted. It is self defeating in content. It is internally inconsistent. Upon reflection, its lack of acceptability to States in general, and its unacceptability to space faring States in particular, should be no surprise to anyone. It has not been and will not be signed and ratified by a substantial number of States. It is, and will remain, of no relevance to the States non-Parties to the Agreement. Thus, it is unlikely ever to have any significance for the use or exploitation of lunar resources. Although some commentators refer to its provisions as if they were equivalent international law to that found in the more generally accepted space agreements formulated through the UNCOPUOS, the 1979 Moon Agreement has never attained such status and it will not in its present form.

In existing international law there are no constraints upon uses of resources beyond the Earth. Signatories to the 1979 Moon Agreement may be bound to intentions that are potentially violating of established international law, but the vast majority of States have declined to adopt that approach. An instructive tour of many existing and anticipatable features of international law related to use of extraterrestrial resources can be found in Sterns, Stine and Tennen's paper dealing with future management of property rights on the moon and other celestial bodies, presented in Beijing, China in October 1996.[17]

Waiting for clarification of international legal rules relating to use of resources beyond the Earth may be self-defeating. The prevailing legal

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[16] Gál, G., *Acquisition of Property in the Legal Regime of Celestial Bodies,* 39 PROC. COLLOQ. L. OUTER SPACE 48 (1997).

[17] P. M. Sterns, G. H. Stine, and L. I. Tennen, *Preliminary Jurisdictional Observations Concerning Property Rights on the Moon and Other Celestial Bodies in the Commercial Space Age,* 39 PROC. COLLOQ. L. OUTER SPACE 50-60 (1997).

regime is permissive and clearly influencable by future action. The longer one waits, the more likely it is that constraints will emerge. The sooner action is taken to demonstrate what can be used and how it can be used, the sooner the international community is likely to move forward to establish appropriate rules to regulate the use and exploitation of extraterrestrial resources.